

GUNNEDAH, Australia

WMO: 957400

Lat: 30.9536S

Lon: 150.2494E

Elev: 263

StdP: 98.20

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.3	0.0	-6.7	2.2	15.1	-4.8	2.6	13.1	9.3	15.7	8.2	15.1	1.0	120	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	38.0	19.8	36.4	19.4	34.7	19.0	22.3	31.0	21.7	30.2	21.0	29.4	5.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.4	24.5	19.4	14.6	24.2	18.6	13.8	23.9	67.0	31.0	64.4	30.3	62.0	29.2	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.7	7.7	DB	-4.2	40.7	1.1	2.4	-5.0	42.4	-5.6	43.8	-6.3	45.1	-7.1	46.8	(4)
(5)			WB	-4.9	23.9	1.0	0.8	-5.6	24.5	-6.1	25.0	-6.7	25.4	-7.4	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	26.3	25.2	22.7	18.6	13.7	10.8	9.7	10.8	14.3	18.6	22.0	24.3	(6)	
	DBStd	6.52	3.11	3.11	2.62	2.93	2.83	2.60	2.52	2.76	3.05	3.37	3.42	3.17	(7)	
	HDD10.0	87	0	0	0	0	4	22	37	22	2	0	0	0	(8)	
	HDD18.3	1081	0	1	2	32	145	225	269	234	126	39	7	1	(9)	
	CDD10.0	3020	505	426	393	257	119	47	27	47	131	266	360	444	(10)	
	CDD18.3	973	247	193	137	38	1	0	0	1	5	46	117	187	(11)	
	CDH23.3	11708	2993	2219	1367	412	23	0	0	10	169	708	1490	2317	(12)	
(13)	CDH26.7	5655	1659	1154	551	88	1	0	0	1	41	249	697	1214	(13)	
(14)	Wind	WSAvg	3.5	4.1	4.3	4.0	3.6	2.9	3.0	2.7	3.0	3.3	3.7	4.0	4.1	(14)
(15) Precipitation	PrecAvg	670	102	66	46	41	56	35	45	40	39	59	65	75	(15)	
	PrecMax	892	301	211	136	150	171	108	169	102	122	144	188	185	(16)	
	PrecMin	341	3	3	0	1	1	2	0	0	1	10	2	1	(17)	
	PrecStd	143	78	52	36	43	45	27	38	25	29	34	47	44	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	40.1	35.5	31.3	25.6	20.8	21.5	24.5	30.4	34.2	38.8	39.2	(19)	
		MCWB	20.4	20.9	19.6	16.9	15.0	13.8	12.4	13.6	14.8	16.3	19.3	19.2	(20)	
	2%	DB	37.9	37.2	33.2	29.1	23.8	19.5	19.3	22.1	27.6	31.5	35.3	36.9	(21)	
		MCWB	20.1	20.2	18.7	17.1	14.3	13.0	12.3	12.6	14.1	16.4	18.0	18.7	(22)	
	5%	DB	36.2	35.1	31.5	27.4	22.3	18.3	17.8	20.2	25.1	29.5	32.9	34.9	(23)	
		MCWB	19.9	19.8	18.4	16.2	13.7	12.2	11.4	11.7	13.5	15.9	17.6	18.5	(24)	
	10%	DB	34.4	33.1	29.9	25.8	20.8	16.9	16.2	18.4	22.8	27.4	30.7	33.0	(25)	
		MCWB	19.7	19.5	18.1	15.7	13.1	11.5	10.7	10.9	13.1	15.4	17.1	18.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	23.7	22.0	20.1	17.4	15.5	15.9	16.2	17.6	19.6	21.2	22.2	(27)	
		MCDB	31.9	33.0	29.4	24.9	21.1	18.3	18.9	21.7	24.2	27.0	31.0	30.8	(28)	
	2%	WB	22.4	22.4	21.1	18.6	16.1	14.2	13.9	14.2	16.3	18.3	20.2	21.2	(29)	
		MCDB	30.9	31.5	28.2	24.7	21.1	17.3	16.7	19.0	22.5	26.1	28.4	29.8	(30)	
	5%	WB	21.7	21.5	20.2	17.5	14.9	13.2	12.4	12.9	15.2	17.4	19.4	20.5	(31)	
		MCDB	30.3	30.3	27.0	24.1	20.0	16.6	16.0	18.1	21.7	25.4	27.4	29.0	(32)	
	10%	WB	21.1	20.7	19.4	16.7	13.9	12.4	11.3	11.7	14.1	16.5	18.7	19.9	(33)	
		MCDB	29.6	29.3	26.4	23.6	19.5	15.8	15.3	17.3	20.9	24.8	26.9	28.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	14.1	13.4	14.5	14.8	12.7	14.0	15.6	16.4	15.9	15.2	14.9	(35)	
		MCDBR	16.6	17.0	15.8	15.8	16.0	14.0	15.7	18.2	20.2	19.1	18.6	18.1	(36)	
		MCWBR	4.7	4.7	5.1	6.0	7.8	8.0	9.3	10.0	9.2	7.4	5.9	5.6	(37)	
	5% WB	MCDBR	13.4	13.9	12.3	12.8	12.9	11.3	12.3	14.1	15.2	15.0	14.0	13.7	(38)	
		MCWBR	4.5	4.5	4.8	5.5	7.0	7.3	8.2	8.4	8.3	6.7	5.2	5.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.357	0.357	0.331	0.312	0.292	0.287	0.282	0.293	0.325	0.341	0.364	0.357	(40)	
	taud		2.527	2.546	2.605	2.619	2.623	2.645	2.642	2.585	2.482	2.473	2.460	2.508	(41)	
	Ebn at Noon		982	964	957	927	899	881	903	933	947	968	969	985	(42)	
	Edh at Noon		112	106	94	83	74	68	72	84	104	113	119	114	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.32	6.62	5.59	4.59	3.56	2.83	3.23	4.17	5.43	6.37	6.86	7.35	(44)	
	RadStd		0.46	0.51	0.41	0.27	0.22	0.24	0.20	0.26	0.38	0.53	0.69	0.53	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (14 neighbors)	+0.58	N/A	N/A	+1.25	N/A	N/A	-16	N/A	+209	+128	(47)

Nomenclature: See separate page